

ALG-07 Floor, Ceiling & Discrete Functions

Student pack - decode interval first, then solve, filter, and check endpoints.

Core router

DISCRETE VALUE -> HALF-OPEN INTERVAL -> CONTINUOUS CONDITION
-> INTEGER FILTER -> ENDPOINT CHECK

RECONNECT

$\text{floor}(x)=n$ means $n \leq x < n+1$. $\text{ceil}(x)=n$ means $n-1 < x \leq n$. The symbols encode half-open intervals; they are not decimal-deletion commands.

Example: $-3 < -2.3 < -2$, so $\text{floor}(-2.3)=-3$ and $\text{ceil}(-2.3)=-2$. Truncation toward zero would give the wrong floor.

Endpoint contract

Floor: include the lower integer, exclude the next. Ceiling: exclude the previous integer, include the upper integer.

Useful identities

For integer k : $\text{floor}(x+k)=\text{floor}(x)+k$ and $\text{ceil}(x+k)=\text{ceil}(x)+k$. Also $\text{ceil}(x)=-\text{floor}(-x)$. Fractional part: $\{x\}=x-\text{floor}(x)$, always $0 \leq \{x\} < 1$.

MAKE SENSE - floor equation

$\text{floor}((2x+1)/3)=4 \rightarrow 4 \leq (2x+1)/3 < 5 \rightarrow 11/2 \leq x < 7$. One integer floor value encodes a full real interval.

MAKE SENSE - ceiling equation

$\text{ceil}((3x-1)/2)=5 \rightarrow 4 < (3x-1)/2 \leq 5 \rightarrow 3 < x \leq 11/3$.

Real first, integer second

If n is an integer and $\text{floor}((n+1)/3)=2$, first solve $5 \leq n < 8$ over the reals, then filter: n in $\{5,6,7\}$.

Integer counting

Integers in $[a,b]$: first $\text{ceil}(a)$, last $\text{ceil}(b)-1$. Integers in $[a,b]$: first $\text{ceil}(a)$, last $\text{floor}(b)$. Always inspect endpoints explicitly.

TRY - H3 to H0 fading

Track A: solve $\text{floor}((x-1)/2)=3$. Attempt first.

H3: $3 \leq (x-1)/2 < 4$. H2: decode the floor interval. H1: which endpoint is strict? H0: solve $\text{floor}((3x+2)/5)=-1$ independently.

Track B: solve $\text{ceil}((x+2)/3)=2$. Attempt first.

H3: $1 < (x+2)/3 \leq 2$. H2: ceiling is left-open/right-closed. H1: decode before algebra. H0: solve $\text{ceil}((2x-5)/4)=-2$ independently.

Track C: integers n with $\text{floor}((n-2)/4)=3$. Attempt first.

H3: $3 \leq (n-2)/4 < 4$, then keep only integers. H2: real interval then Z-filter.

H1: do not count before the interval is correct. H0: count integers satisfying $\text{ceil}(n/5)=3$.

DIAGNOSE

Wrong starts: floor=truncation; $\text{floor}(f(x))=k$ replaced by $f(x)=k$; both endpoints made inclusive; integer filtering done before interval solving; general inequality machinery used when simple interval decoding is enough.

ADOPT

1) floor -> $[n,n+1)$. 2) ceiling -> $(n-1,n]$. 3) negative input: use order. 4) integer shifts pass through. 5) $\text{ceil}(x)=-\text{floor}(-x)$. 6) $\{x\}=x-\text{floor}(x)$. 7) decode -> solve -> filter -> endpoint check.

First-Step Reference

$\text{floor}(A)=n \rightarrow n \leq A < n+1$. $\text{ceil}(A)=n \rightarrow n-1 < A \leq n$.

Negative input -> locate between consecutive integers, not truncation.

Integer shift -> use translation identity. Integer target -> solve real interval, then intersect Z. Fractional part -> $x-\text{floor}(x)$.

Mandatory contrasts

Floor vs truncation; floor vs ceiling; real interval vs integer solution set; floor equation vs ordinary equation; included vs excluded endpoint. Every contrast changes the first move or endpoint check.

Validated anchors

IOQM-2024-Q21: intersect two floor-generated integer intervals with digit structure; independent answer 91. IOQM-2024-Q26: set $n=\text{floor}(x)$, use x in $[n,n+1)$, and test endpoint feasibility; floor values 16 and 17 sum to 33.

Recognition & First-Line Lab

1. Evaluate $\text{floor}(3.8)$ by locating it between consecutive integers.

2. Evaluate $\text{floor}(-3.8)$ without truncation.

3. Evaluate $\text{ceil}(-3.8)$.

4. $\text{floor}(x)=5$. Write the encoded interval.

5. $\text{ceil}(x)=-2$. Write the encoded interval.

6. $\text{floor}((2x+1)/3)=4$. Write the first inequality chain.

7. $\text{floor}(x)<3$. Write an equivalent ordinary inequality.

8. When can $\text{floor}(x)=x$?

9. Write the defining first line for the fractional part of -2.3 .

10. Count integers in $[-2.4,3.1)$ by identifying first and last candidate.

11. Simplify $\text{floor}(x+7)$ without casework.

12. Express $\text{ceil}(x)$ using floor and reflection.

13. $\text{floor}(x)=2$ and x is integer. Write interval before filtering.

14. $\text{floor}(x^2)=3$. Write the first double inequality.

15. $\text{floor}(x)=\text{ceil}(x)$. State the structural condition to test.

16. For $\text{floor}(x)=4$, identify included and excluded endpoints.

Practice - F0/F1

1. Evaluate $\text{floor}(4.6)$, $\text{floor}(-4.6)$, $\text{ceil}(4.6)$, $\text{ceil}(-4.6)$.
2. Translate $\text{floor}(x)=n$ and $\text{ceil}(x)=n$ into inequalities.
3. Find $\{3.25\}$ and $\{-2.3\}$.
4. Verify $\text{ceil}(x)=-\text{floor}(-x)$ for $x=2.4$ and $x=-2.4$.
5. Solve $\text{floor}(x)=4$.
6. Solve $\text{ceil}(x)=4$.
7. Solve $\text{floor}((x+2)/3)=2$.
8. Solve $\text{floor}(-x)=3$.
9. If $3 < x < 4$ and $\{x\}=0.25$, find x .
10. Solve $\text{floor}(2x+1)=5$.
11. Solve $\text{ceil}(3x-2)=4$.
12. Solve $\text{floor}(x)=\text{ceil}(x)$.
13. Solve $\text{floor}(x)+\text{ceil}(x)=7$.
14. Classify $\text{floor}(x)+\text{floor}(-x)$ for integer and noninteger x .
15. Solve $\text{floor}(x^2)=4$.

Practice - F3/F4 Transfer

16. How many integers lie in $[-3.7, 5.2)$?
17. How many integers k satisfy $\text{floor}((k+1)/3)=2$?
18. Solve simultaneously $\text{floor}(x)=2$ and $\text{floor}(2x)=5$.
19. Explain why $\text{ceil}(x)=-\text{floor}(-x)$ follows from interval definitions.
20. Characterize x such that $\text{floor}(x+1/2)=\text{floor}(x)$.
21. Can $\text{floor}((x-1)/2)=\text{floor}((x+1)/2)$ ever hold? Prove without case enumeration.
22. Solve $\text{floor}(x)+\text{floor}(x+1/2)=5$.
23. Find x if $\text{floor}(x)=3$ and $\{x\}=0.7$.
24. If $\text{floor}(x)=-4$, state the full possible interval and explain truncation risk.
25. Find all integers n satisfying $\text{floor}(n/4)=3$.
26. How many integers n satisfy $\text{ceil}(n/5)=3$?
27. For $x>0$, solve $\text{floor}(1/x)=2$.
28. Prove $\text{floor}(x+k)=\text{floor}(x)+k$ for integer k .
29. WHY-NOT: why does $\text{floor}(x)=2$ not allow replacing it by $x=2$?
30. A gate is active for t in $[2.3, 8.0)$. How many integer timestamps occur?

H0 Mastery Test

1. $\text{floor}((3x-2)/4)=5$. Write only the first inequality chain.
2. $\text{ceil}((x+3)/2)=-1$. Write only the first inequality chain.
3. $\text{floor}(x^2)=7$. Write only the first double inequality.
4. How many integers n satisfy $-4.2 \leq n < 3.1$? Identify first and last integer.
5. Solve $\text{floor}((2x+1)/3)=4$.
6. Solve $\text{ceil}((3x-1)/2)=5$.
7. Solve $\text{floor}(x)+\text{ceil}(x)=-5$.
8. Solve $\text{floor}(x^2)=9$.
9. Find all integers k satisfying $\text{floor}((k-2)/4)=3$.
10. Find the fractional part of -2.75 .
11. Solve $\text{floor}(x)=\text{ceil}(x)$.
12. Characterize all x satisfying $\text{floor}(x+0.4)=\text{floor}(x)$.
13. Solve $\text{floor}(x)=4$ and $\text{ceil}(x)=4$; explain endpoint difference.
14. Compare $\text{floor}(-2.3)$ with truncation toward zero; identify controlling definition.
15. Seat labels are integers. How many labels lie in $[12.4, 20.0]$?
16. A learner solves $\text{floor}(2x)=3$ by writing $2x=3$. Explain defect and give full solution.

Mastery reflection

For any three items: What interval was encoded? Which endpoint was strict? Was an integer filter required? What wrong move was tempting? What final endpoint/domain check did you perform?